IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-108

1. Topic (12 words max)

Smart Multi-Sensor Wearable for Real-time Elderly Health Anomaly and Fall Detection

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	R26-IT-108
Year	2026

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Nature/Gravity of the Problem (User Persona):

The primary target audience is the independent elderly population (aged 65+) living alone or in assisted living facilities. This demographic is critically vulnerable to "silent" physiological deterioration (such as asymptomatic hypoxia or sepsis) and physical trauma (falls). Furthermore, non-verbal seniors often suffer from undetected pain or anxiety (emotional distress), which cannot be detected by standard cameras or fall detectors.

- ***The Gravity:*** *Falls are the leading cause of injury-related death among seniors. Conditions like "silent hypoxia" and untreated chronic stress/pain lead to rapid physical decline and*

6. Brief description of Existing Research and Systems (300 words max)

Summary of Existing Products/Systems:

Commercial market leaders (Apple Watch, Fitbit) offer general health monitoring but are prohibitive in cost and complexity. Critically, their Fall Detection algorithms are generalized for active adults, leading to false alarms. Most consumer devices lack specific "medical fusion" capabilities, such as correlating Temperature with Heart Rate for sepsis, or monitoring Galvanic Skin Response (GSR) for stress/pain detection.

Summary of Existing Research:

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

Addressing the Gaps:

We propose a Multi-Modal Intelligent Health Band utilizing Sensor Fusion and TinyML. This system processes information directly on the microcontroller. It addresses Contextual Blindness by correlating Motion, Heart Rate, and Skin Conductance (GSR) to distinguish between exercise, distress, and pain. It addresses Latency by performing anomaly detection on the device itself.

System Perspective (Input → Process → Output):

- ***1. Input Layer (Sensors):***

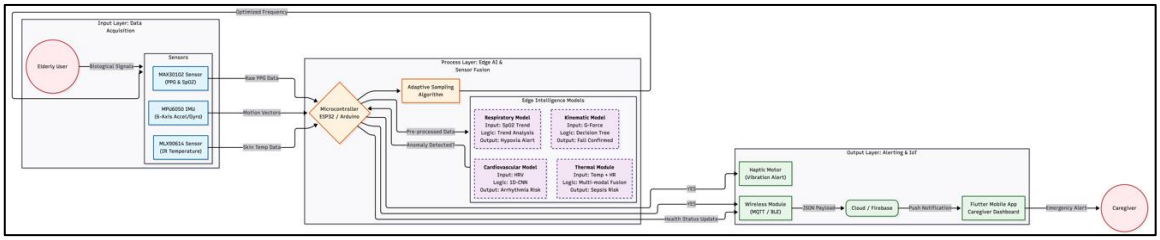


Figure 1: Conceptual Diagram

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The team will research biomedical signal processing, focusing on PPG noise reduction and Electrodermal Activity (EDA) analysis. We will consult medical standards to determine thresholds for hypoxia, sepsis, and physiological stress responses in the elderly.

Gaining Technical Knowledge:

We will undertake self-directed learning in Embedded Machine Learning (TinyML) using TensorFlow Lite for Microcontrollers. We will learn to interface analog sensors (GSR) and digital I2C sensors with the ESP32.

Data Collection Strategy:

We will use established open-source datasets to train our ML models:

1. **MIT-BIH Arrhythmia Database:** For arrhythmia detection.
2. **SisFall / KFall Dataset:** For fall detection.
3. **MIMIC-III Clinical Database:** For sepsis/temperature correlation.
4. **WESAD (Wearable Stress and Affect Detection):** This dataset contains multimodal data (ECG, EDA/GSR, Temp) labeled for "Stress," "Amusement," and "Baseline," which we will use to train the stress/pain monitoring model.

Ethical Clearance:

9. Objectives and Novelty

Main Objective

To design and develop a privacy-centric, multi-sensor wearable system that utilizes Edge Computing (TinyML) to detect cardiac, respiratory, thermal, kinematic, and neurological anomalies directly on the ESP32 microcontroller. Unlike traditional cloud-based systems, this approach processes confidential health data locally on the device to predict anomalies in real-time, ensuring data privacy and eliminating the security risks associated with transmitting raw medical signals over networks.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Abeysinghe S.A.M.M. *IT22081148*	***Cardiovascular Anomaly Detection System***	*1. Integrate MAX30102 sensor and preprocess PPG signals.* *2. Train 1D-CNN model on MIT-BIH dataset.* *3. Deploy model to microcontroller for real-time arrhythmia classification.*	*Implementation of Edge-based Arrhythmia classification (Normal vs. Abnormal) directly on the wearable, ensuring data privacy and eliminating cloud latency during cardiac events.*
Lakshan N.K.T. *IT22196156*	***Kinematic Fall & Immobility Detection***	*1. Interface MPU6050 & calibrate 6-axis data.* *2. Implement "Impact + Immobility" logic.* *3. Train Decision Tree model on SisFall dataset to reduce false positives.*	*A Two-Stage Verification Algorithm that correlates high-G impact with post-event orientation vectors to distinguish actual falls from daily forceful movements (like clapping).*

Dissanayaka D.M.S.K. *IT22291196*	***Respiratory & Thermal Warning System***	*1. Integrate MLX90614 & calibrate SpO2 readings.* *2. Develop "Sepsis Risk" algorithm combining Temp + HR.* *3. Implement trend analysis for Silent Hypoxia.*	*Multi-modal Risk Correlation: Unlike standard thermometers, this module cross-references Temperature with Heart Rate trends to predict systemic infection (Sepsis) onset before it becomes critical.*
Peshala E.A.P. *IT22146410*	***Neurological Stress & Pain Monitoring***	*1. Interface **GSR Sensor** to measure skin conductance.* *2. Train model on **WESAD dataset** to detect stress states.* *3. Develop logic to distinguish "Stress" from "Exercise".*	***Objective Pain/Stress Detection:** Introducing **Affective Computing** to elderly care to detect silent pain or anxiety in non-verbal patients by fusing GSR and HR data.*

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	X	No	

b) The proposed topic exhibits novelty.

Yes	∧	No	

c) The student group can successfully execute the proposed project.

Yes	X	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	X	No	

e) Any other comments:

Highly recommended

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Nelum	Amaraseya	
Co-Supervisor	Mr.	Samadhi	Rathnayake	
External Supervisor	Prof	S A M	Kularatne	
Summary of external supervisor's (if any) experience and expertise				

11-1-2026

Prof. S.A.M. Kularatne
MBBS, MD, MRCP, FRCP (Lond), FCCP, FNASSL
DSc (honoris causa)-Peradeniya
Consultant Physician
Emeritus Professor of Medicine
University of Peradeniya
Sri Lanka